

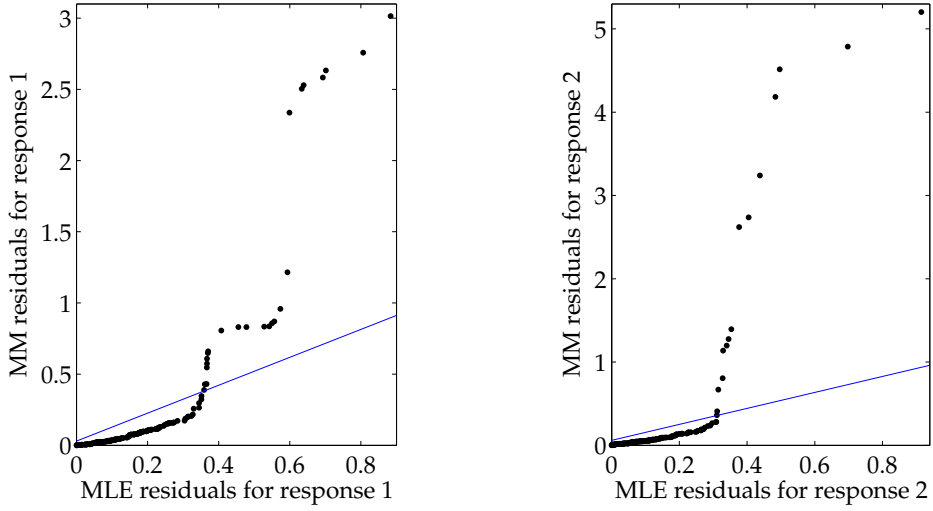


Figure 6: QQ-plots of sorted absolute residuals of MM-estimates vs sorted absolute residuals of MLE for each component of the response. Left plot corresponds to  $P_2O_5$  (first component) and right to  $PbO$  (second component).

| Criterion     | MLE   |       | $\tau$ -estimate |       | MM-estimate |       | MM-univariate |       |
|---------------|-------|-------|------------------|-------|-------------|-------|---------------|-------|
| Component     | 1     | 2     | 1                | 2     | 1           | 2     | 1             | 2     |
| MSE           | 0.081 | 0.051 | 0.351            | 0.806 | 0.340       | 0.682 | 0.354         | 0.762 |
| $\tau$ -scale | 0.044 | 0.022 | 0.005            | 0.007 | 0.008       | 0.006 | 0.005         | 0.006 |

Table 5: Mean square error (MSE) and  $\tau$ -scale of the prediction errors of the MLE, multivariate MM-estimate,  $\tau$ -estimate and univariate MM-estimate for each component of the response, computed by cross-validation.

square root quantiles of the chi-squared distribution with  $q$  degrees of freedom. For ease of comparison, the center panel shows the MM's QQ-plot truncated to the size of the MLE's QQ-plot. The latter shows a very good fit of the norms to the chi-squared distribution, and therefore points out no suspect points, while the MM-estimate's QQ-plot indicates some 30 possible outliers, i.e. about 16% of the data.

The MLE's norms are in general smaller than the MM-estimate's norms, but this does not mean that the former gives a better fit, since here the residuals are normalized by the respective estimated residual dispersion matrices  $\hat{\Sigma}_0$ . Figure